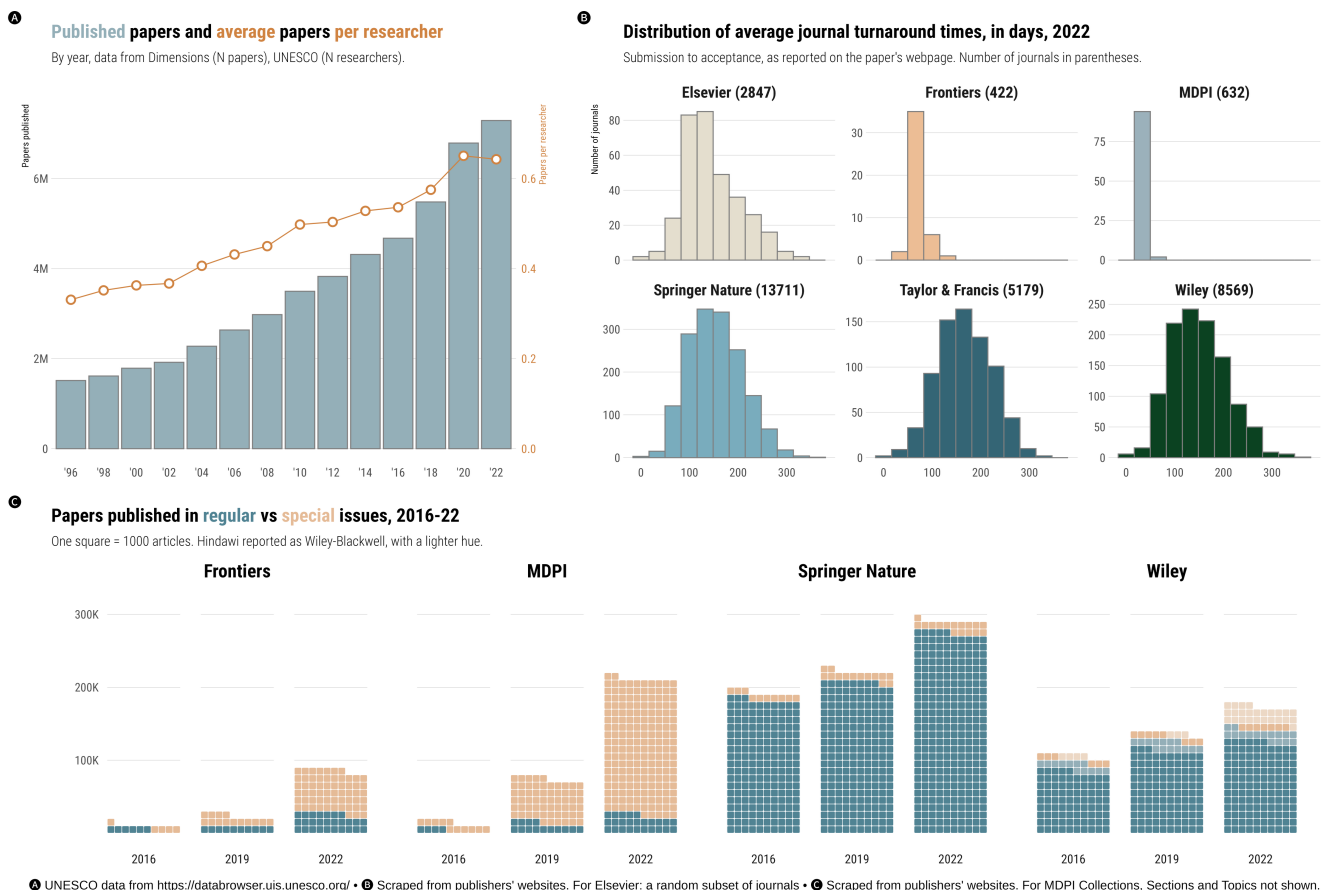


limited progress whereas core scholarly practices – such as reading, reflecting and engaging with others’ contributions – is de-prioritized. What looks like productivity often masks intellectual exhaustion built on a demoralizing, narrowing scientific vision. Reforms – such as recognizing or compensating peer review, improving evaluation metrics, and rewarding quality over quantity are vital – but they do not address the drive for productivity that stems from for-profit business models.

Figure 1: Concerning Aspects of Scientific Publishing.



1.3 Trust

Maintaining integrity, equity and efficiency within this remarkable growth has become increasingly difficult (11). The integrity of scholarly publishing, and its role as a trusted provider of knowledge, rests on authors carrying out research responsibly and writing it up accurately, and on editors and reviewers scrutinising the words, data and images with care. Historically, the space constraints in print-on-paper journals incentivized publishers’ and editors’ vigilance in this process (if not necessarily guaranteeing quality). The move to digital publishing has removed that constraint and changed publishers’ (if not researchers’) need for vigilance. Publishers can now make money from publishing quantity as well as quality, whilst still appearing to follow previous conventions of rigorous pre-publication gate keeping.

The proliferation of journals under the Nature brand is one response to this opportunity, as are ‘cascade’ policies in which publishers seek to place rejected papers in other journals from which they make money. Brands like MDPI, Hindawi (now part of Wiley), Discover (part of

Springer Nature) and Frontiers have been particularly adept at seizing on the opportunity presented by a loss of print-on-paper constraints, producing numerous special issues and soliciting authors to contribute to them (Figure 1C and S2).

There is evidence of publishers' new business models affecting editorial independence. Editorial boards have resigned because of the interference of commercial interests, such as 'cascade policies' (S2). The striking lack of variation in turnaround times that some brands achieve across multiple disciplines and hundreds of journals (Figure 1B) suggests a shift from academic editorship to a commercial management of the publishing process.

Such strategies have been seen to backfire, with peer review rings, paper mills, and AI-generated fraud triggering mass retractions and raising concerns about editorial oversight (S2). Some journals have been hijacked (12), and entire brands, such as Hindawi, shut down due to the damage caused by paper mills. These are not isolated failures, but symptoms of a research system powerfully shaped by for-profit incentives. The scrutiny and selectivity that peer-reviewed journals were once believed to stand for becomes harder to sustain. Peer review was never flawless, but now the conventional peer review system, based on anonymous and confidential judgements informing decisions to publish (or not), no longer inspires the trust that it once did. The speed and volume of current publishing activity all come at the expense of rigor.

1.4 Control

While academics often retain editorial oversight of journals, even when published by commercial firms, they rarely direct marketing policies, set financial policy, or control brands. Back in the 1960s, a group of UK journal editors had tried to insist that scientific journals should remain under full academic control even when published by commercial firms (1). But this proved difficult to implement in an age of increasing demands on academic time, and the increasing complexity of the journal publishing industry.

Yet it is not just the journals themselves that are increasingly controlled by commercial firms: so too are the processes of research evaluation. Some firms are now involved in both journal publishing and compiling the data for quantitative research assessment. Clarivate (now private-equity owned) publishes the Journal Impact Factor via its Journal Citation Reports, a set of prestigious indexes governed by opaque decision-making. Elsevier owns Scopus and an expanding arsenal of analytic platforms that capture the entire research lifecycle. Despite calls for reform (13), the continued reliance on these metrics for scientific governance, evaluation and career progression has skewed decision making and incentivized gaming of publication metrics for prestige.

Just as prestige metrics are privately controlled, so too are the institutions tasked with safeguarding publishing integrity – further distancing them from the control of the academic community. The Committee on Publication Ethics (COPE), founded in the 1990s by concerned editors specifically to address misconduct, now includes over 14,000 journals and 100 publishers who act as judge and jury of their own trials.

More broadly, control over scholarly publishing remains concentrated in the Global North with Elsevier, Clarivate, and COPE headquartered in the Netherlands, the US and the UK. This geographic dominance only entrenches Northern norms as the global standard of 'research excellence', thereby marginalizing research published in languages other than English, with regional relevance, or in non-commercial venues (8). Platforms such as SciELO, Redalyc, Latindex or African Journals Online offer quality, community-governed alternatives and use clear criteria and local oversight to maintain standards and detect

spurious content. Yet, these regional models remain undervalued in evaluation systems shaped by Northern priorities, perpetuating global inequities in defining research quality.

2. Stopping the drain

Back in the 1950s, when for-profit journal publishing was just getting going, British scientific leaders predicted that ‘the moment commercial gain began to dominate this field, the welfare of the scientific community would suffer’ (1). They were right – and it is past time that we acted. The first step we must take is to recognize the seriousness of the problems that scientific publishing’s engagement with for-profit publishing have become.

The four-fold drain means that for-profit publishing and the prestige economy it sustains no longer serve the interests of the research community. The aims of different actors in scholarly communication remain fundamentally misaligned. This means that we cannot work with commercial publishers to produce system-wide reform. With profit margins above 30%, the temptation will always be to pursue revenue over science.

The futility of working with commercial publishers is clear from the multiple failed attempts to do so. Since the serials crisis emerged in the 1980s (14), decades of attempted remedies and over 25 years of Open Access initiatives have applied band-aids to a hemorrhage. Funder policies intended to counteract this have often backfired. For example, in the UK, the system of APC block grants normalized high publishing charges and accelerated the growth of hybrid Open Access. Targets set by Coalition S and others aiming to form fully Open Access journals have not been met. The one constant amidst all these reforms are publishers’ profits.

Our analysis also suggests that technological advances will not help. Innovations in AI might improve the efficiency of editorial processes or perhaps assist with some form of preliminary review. This will speed things up. It will generate more papers, and thus more profits for publishers. It will not increase researchers’ control over standards. We do not need technology to make commercial publishing quicker. We need to change the structures and incentives governing publishing.

Effective change will thus have to alter the structures – the incentives in, and ownership of, scientific publishing. This will require the interventions of the powerful organizations which shape the evaluation of research and determine its funding. Public funding agencies, foundations and universities have means to act in ways which could transform the current publishing ecosystem. Encouragingly, some are beginning to act more decisively. Some funders already demand all work be preprinted, while others, such as the US NIH, have capped or disallowed the use of funds to pay the APCs of special issue articles (15, S2). It is, however, essential that such action should be based on a strategic vision for a future system, something that needs greater discussion and debate.

We propose that scholarly publishing needs to be re-communalized. Universities, libraries, funders and other members of the academic community need to build a system that is community-led and managed, and which works to further research and education. This should involve active support for federated open infrastructures (such as LA Referencia), and investment in community-based publication platforms and non-commercial journals; many of which exist outside the anglo-american space (Érudit, OpenEdition, SciELO). Encouraging innovation in scholarly communication systems, rather than simply reproducing old models is critical, and might include more work in approaches like ‘publish, review, curate’, which re-conceptualizes the publication and peer review process. Genuine academic oversight needs to be part of this. Policies mandating good practice in disseminating research, such as making work Open Access in affordable ways, are also essential.

The vision of re-communalization may require disruptive change (depending on the political economy of scientific publishing in different world regions) and, as such, cannot be achieved overnight. Major actors in the research environment may work in concert to dismantle the system that currently grants for-profit corporations control over science. However, coordination can be difficult to achieve, and waiting to do so will delay action where it is urgently needed. We, therefore, urge that leading actors in the research community should do just that – lead. Where there are opportunities to act, we recommend that researchers, funders and others take the initiative, and that crucially they do so in ways that further the long-term vision of re-communalization.

Private organisations might provide services to the research community in a future system, but only in ways that do not involve extractive profits. Thus, in the immediate term, decisive intervention in the market should be considered. Competition authorities need to take seriously the dysfunctional nature of the current market and intervene to curb profiteering behaviours and create more meaningful competition, in a manner similar to action that led to the failed merger between Reed Elsevier and Wolters Kluwer almost three decades ago. Radical action by the funders like taking shares in large publishing organizations and exerting pressure from within could also be given consideration – shareholder pressure is a proven lever in changing corporate strategy.

At the same time, a major challenge is to change the incentives and reward structures that shape researchers' publishing behaviours. This means reconfiguring the prestige economy, arguably the most difficult but most crucial task. The current prestige system incentivizes publishing in highly cited journals controlled by commercial publishers. Initiatives like DORA, CoARA, FOLEC and the Barcelona Declaration have contributed to the public discussions on issues of research assessment and reassessing prestige. But they need stronger support from powerful funders, and their advocacy needs more directly to reward research published in community-owned journals and disincentivize publishing in commercially run journals. Again, the funders – governments, universities and foundations - have considerable power here to reward publication outside traditional venues, such as provided by Latin American models, and discourage publishing in commercial outlets.

The costs of inaction are plain: wasted public funds, lost researcher time, compromised scientific integrity and eroded public trust. Today, the system rewards commercial publishers first, and science second. Without bold action from the funders we risk continuing to pour resources into a system that prioritizes profit over the advancement of scientific knowledge.

References:

1. A. Fyfe, Self-help for learned journals: Scientific societies and the commerce of publishing in the 1950s. *History of Science* **60**, 255-279 (2022).
2. L.-A. Butler, L. Matthias, M.-A. Simard, P. Mongeon, S. Haustein, The oligopoly's shift to open access: How the big five academic publishers profit from article processing charges. *Quantitative Science Studies* **4**, 778-799 (2023).
3. S. Haustein *et al.*, Estimating global article processing charges paid to six publishers for open access between 2019 and 2023. <https://arxiv.org/abs/2407.16551>, (2024).
4. V. Larivière, S. Haustein, P. Mongeon, The Oligopoly of Academic Publishers in the Digital Era. *PLoS ONE* **10**, e0127502 (2015).
5. T. C. Bergstrom, P. N. Courant, R. P. McAfee, M. A. Williams, Evaluating big deal journal bundles. *Proceedings of the National Academy of Sciences* **111**, 9425-9430 (2014).